

2-STAGE OTA DESIGN ASSIGNMENT

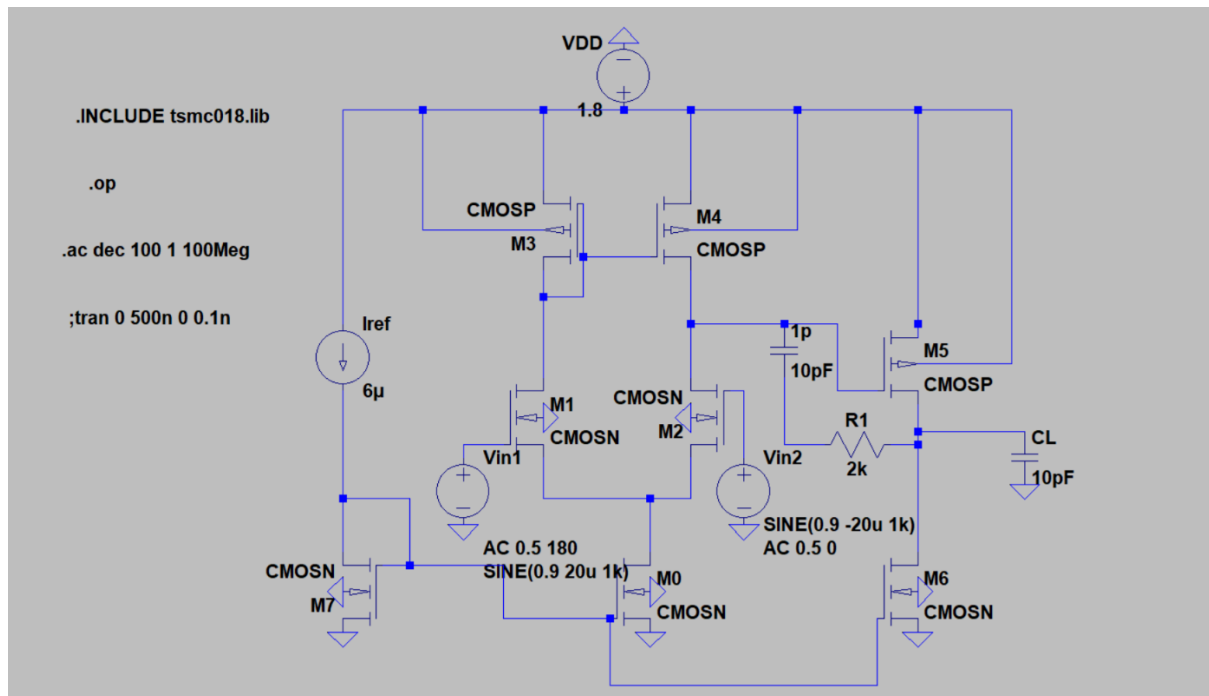
NAME: B.SRISHANTH

ROLL NO: 240102023

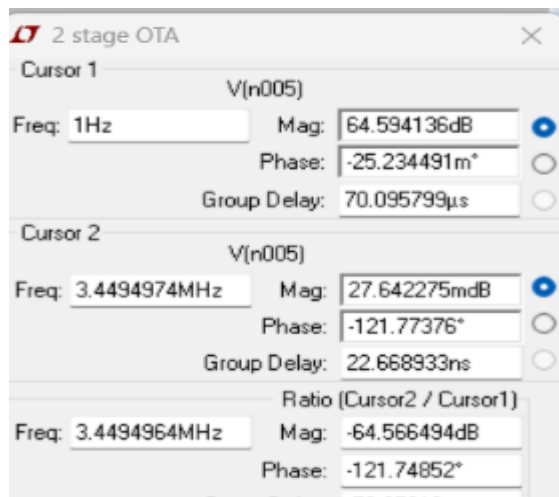
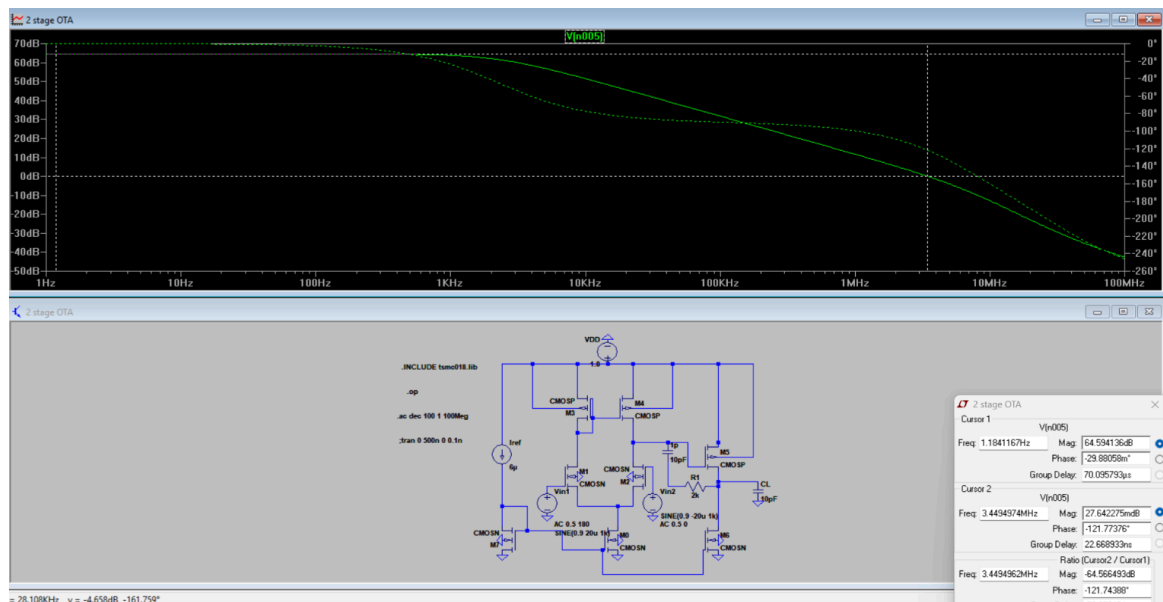
Considered values

$C_l = 10\text{pF}$, $C_c = 10\text{pF}$, $R = 2\text{k ohm}$

OPEN LOOP CIRCUIT CONFIGURATION



OPEN LOOP FREQUENCY RESPONSE

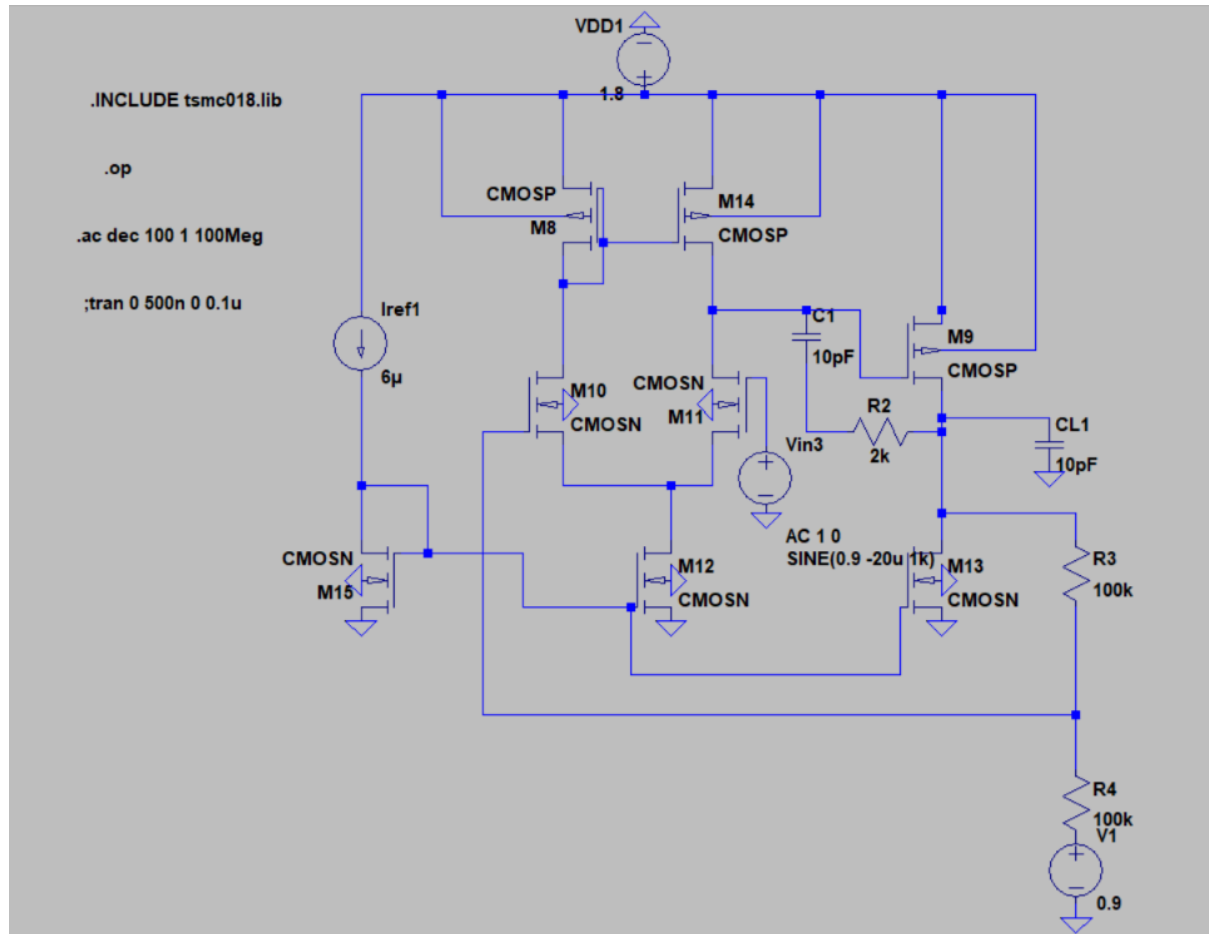


Frequency Response Characteristics of open loop circuit:

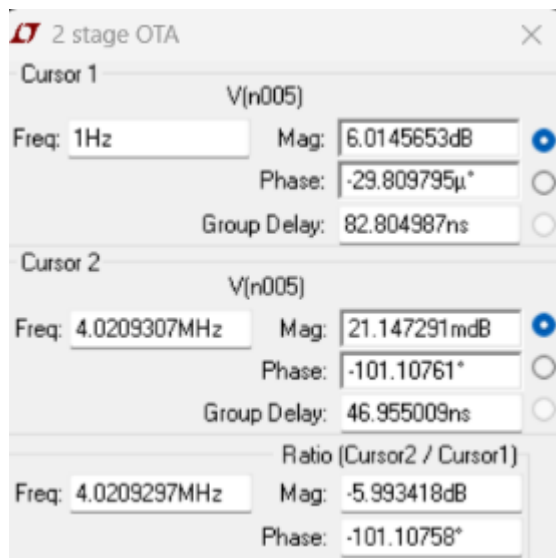
Low Frequency Gain = 64.594 dB

Phase Margin = $180 - 121.748 = 58.252$ degrees

CLOSED LOOP CIRCUIT CONFIGURATION



FREQUENCY RESPONSE

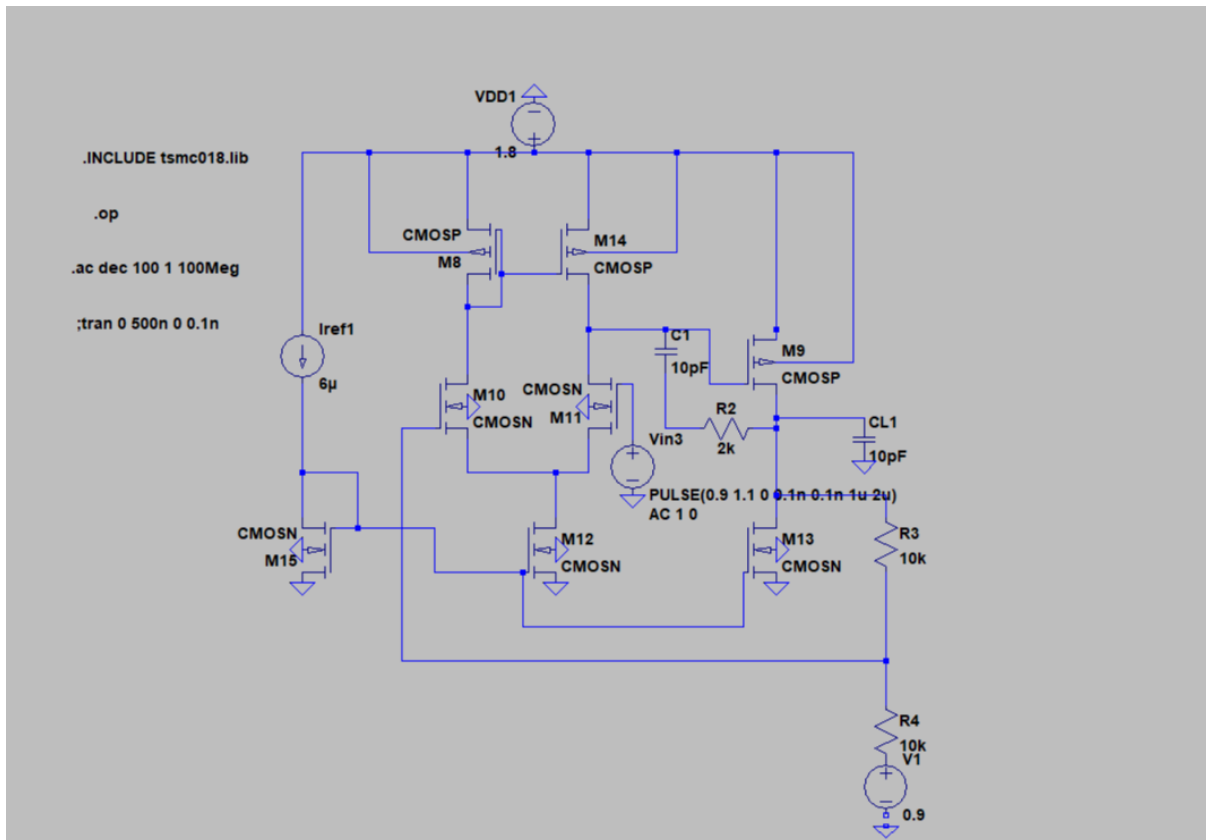


Frequency Response Characteristics o closed loop circuit:

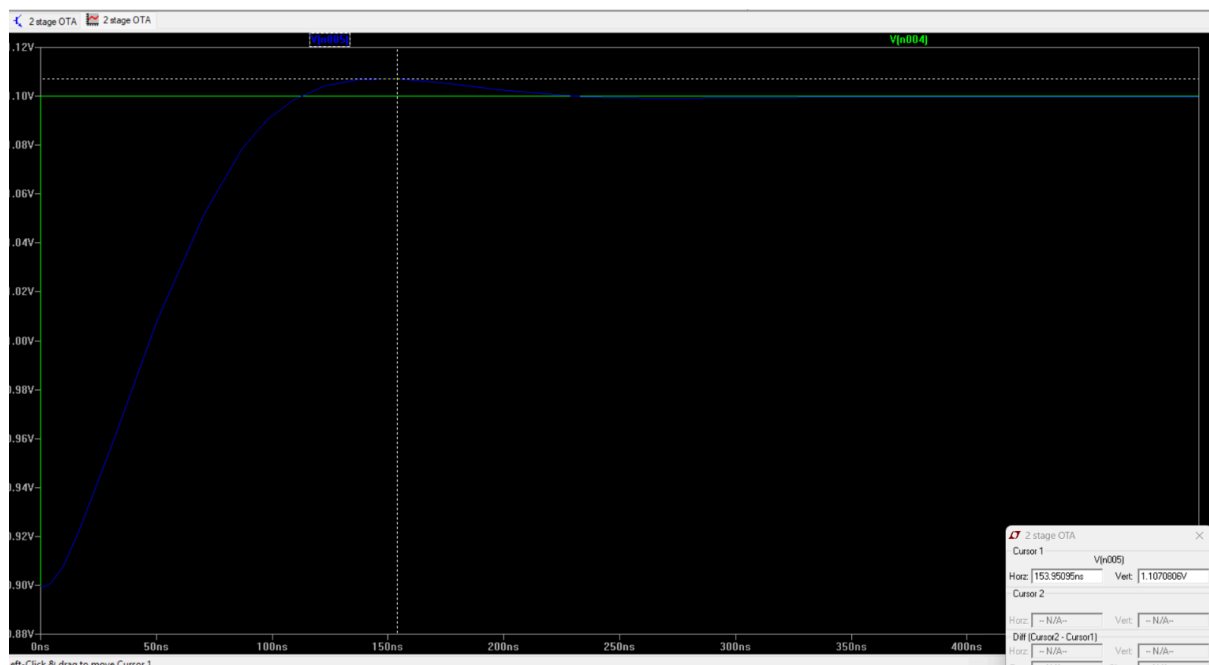
Low Frequency Gain = 6.015 dB

Phase Margin = $180 - 101.107 = 78.893$ degrees

CLOSED LOOP STEP RESPONSE CIRCUIT:



STEP RESPONSE:



The GREEN plot represents the input signal with an offset of 0.9, a steady-state value of 1.1, and an amplitude of 0.2. The BLUE plot shows the output signal